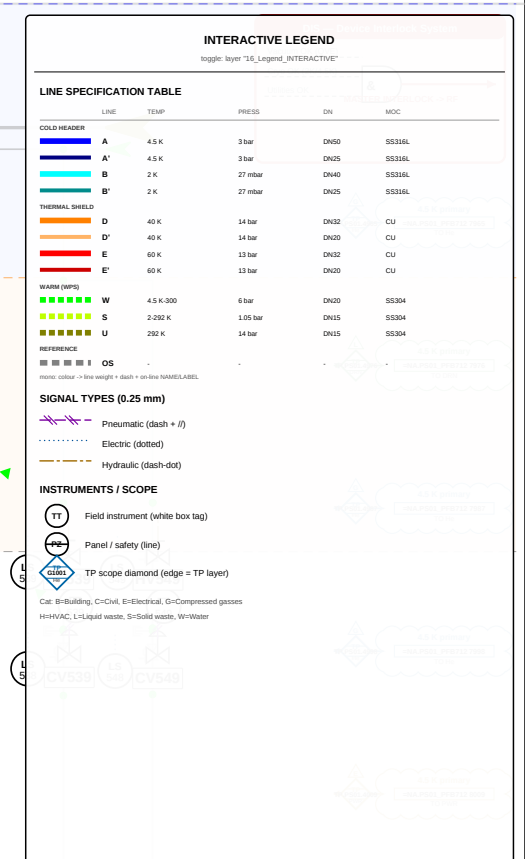


**COLD HEADER - Line A (4.5 K) / Line B (2 K)**



## INTERACTIVE LEGEND

toggle: layer "16\_Legend\_INTERACTIVE"

### LAYER SPECIFICATION TABLE

| LINE                                                                                   | TEMP      | PRESS    | DN   | MOC    |
|----------------------------------------------------------------------------------------|-----------|----------|------|--------|
| <b>COLD HEADER</b>                                                                     |           |          |      |        |
|  A  | 4.5 K     | 3 bar    | DN50 | SS316L |
|  A' | 4.5 K     | 3 bar    | DN25 | SS316L |
|  B  | 2 K       | 27 mbar  | DN40 | SS316L |
|  B' | 2 K       | 27 mbar  | DN25 | SS316L |
| <b>THERMAL SHIELD</b>                                                                  |           |          |      |        |
|  D  | 40 K      | 14 bar   | DN32 | CU     |
|  D' | 40 K      | 14 bar   | DN20 | CU     |
|  E  | 60 K      | 13 bar   | DN32 | CU     |
|  E' | 60 K      | 13 bar   | DN20 | CU     |
| <b>WATER (PERT)</b>                                                                    |           |          |      |        |
|  W  | 4.5 K-300 | 6 bar    | DN20 | SS304  |
|  S  | 2-292 K   | 1.05 bar | DN15 | SS304  |
|  U  | 292 K     | 14 bar   | DN15 | SS304  |
| <b>REFERENCE</b>                                                                       |           |          |      |        |
|  OS |           |          |      |        |

note: colour -> line weight + dash + on-line NAME/LABEL

### SIGNAL TYPES (0.25 mm)

|                                                                                     |                      |
|-------------------------------------------------------------------------------------|----------------------|
|  | Pneumatic (dash + /) |
|  | Electric (dotted)    |
|  | Hydraulic (dash-dot) |

### INSTRUMENTS / SCOPE

|                                                                                     |                                    |
|-------------------------------------------------------------------------------------|------------------------------------|
|  | Field instrument (white box tag)   |
|  | Panel / safety (line)              |
|  | TP scope diamond (edge = TP layer) |

Car: B=Building, C=Cell, E=Electrical, G=Compressed gasses

H=HVAC, L=Liquid waste, S=Solid waste, W=Water

|                                                                        |  |                             |           |         |                              |           |                                                                                                                                                                                                                                                                                                                                                                                       |          |      |
|------------------------------------------------------------------------|--|-----------------------------|-----------|---------|------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|
| <div>CONSULTANT</div> <div>Mott MacDonald</div> <div>Bristol, UK</div> |  | CLIENT                      | REVISIONS |         |                              | APPROVALS | <div>MINERVA CryoCell - SCK CEN</div> <div>QCELL / LB - Instrumentation &amp; Control</div> <div>DWG No. =NA.P501.PFB712</div> <div>MMO Proj. 411066</div> <div>Scale NTS</div> <div>Sustainability S2 - FOR ACCEPTANCE</div> <div>Security RESTRICTED</div> <div>Size / Std A3 / ISO10628 / ISA-5.1</div> <div>Sheet Sheet2 of 2 - Instrumentation</div> <div>Variant STANDARD</div> |          |      |
|                                                                        |  | SCK CEN                     | Rev       | Date    | Description                  | App'd     |                                                                                                                                                                                                                                                                                                                                                                                       | Designed | QSYS |
|                                                                        |  | Boeratang 200, 2450 Mol, BE | C1        | 2026-06 | v3 refinement - layers/legib | ACR       |                                                                                                                                                                                                                                                                                                                                                                                       | Drawn    | ACR  |
|                                                                        |  | MYRRHA / MINERVA Phase 1    | B1        | 2026-05 | v2 split sheets              | ACR       |                                                                                                                                                                                                                                                                                                                                                                                       | Checked  | ACR  |
|                                                                        |  |                             | A1        | 2026-04 | First issue                  | ACR       |                                                                                                                                                                                                                                                                                                                                                                                       | Approved | PM   |